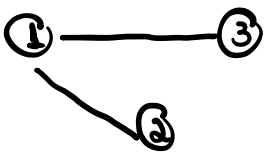
6.1.1 b F

b)

	1	2	3
1	1	1	0
2	0	0	0
3	1	0	1

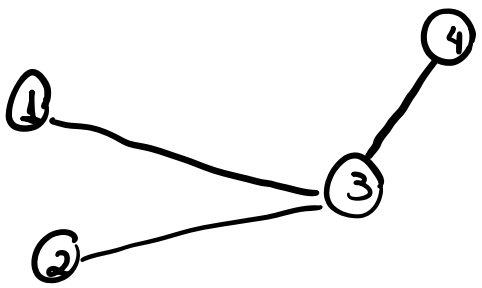
Ordered Pairs:
(1,1), (1,2), (3,1), (3,3)



f)

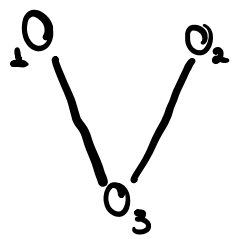
	1	2	3	4
1	0	0	1	0
2	0	0	1	0
3	0	0	1	0
4	0	0	1	0

Ordered Pairs:
(1,3), (2,3), (3,3), (4,3)



6.1.2 b F

b)

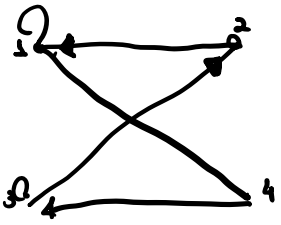


MATRIZ

$$M = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$R = \{(1,1), (1,3), (2,2), (2,3)\}$

f)



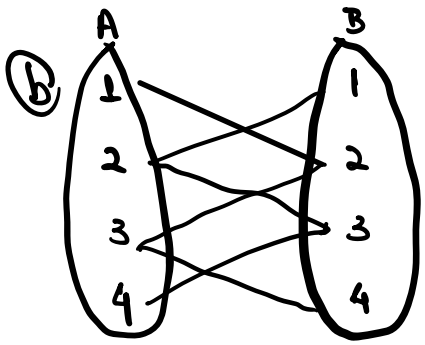
MATRIX

$$M = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$R = \{(1,1), (1,4), (2,1), (3,3), (4,3), (3,2)\}$

6.1.3 bd

b)

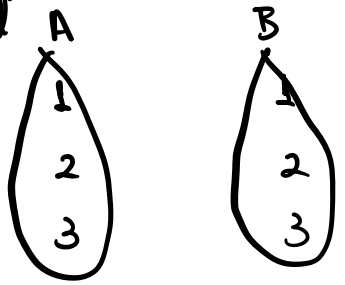


MATRIZ

	1	2	3	4
1	0	1	0	0
2	1	0	1	0
3	1	1	0	1
4	0	0	1	0

4x4

Q



MATRIX

	1	2	3
1	0	0	0
2	0	0	0
3	0	0	0

3×3

6.2.1 bd

(b) domain & Relation is $\subseteq \mathbb{R}$, for $x, y \in \mathbb{R}$

$x \mathrel{E} y \leftrightarrow x \leq y$

(i) E is reflexive

As we know $x = x \quad \forall x \in \mathbb{R}$

$\rightarrow x \leq x$

$\rightarrow x \mathrel{E} x \quad \forall x \in \mathbb{R}$

(ii) E is Anti Symmetric

let $x \mathrel{E} y \rightarrow x \leq y$ — (A)

$y \mathrel{E} x \rightarrow y \leq x$ — (B)

$\rightarrow x = y$ [From (A & B)]

(iii) E is Transitive

let $x \mathrel{E} y \quad y \mathrel{E} z, \ \forall x, y, z \in \mathbb{R}$

$\rightarrow x \leq y \quad \& \ y \mathrel{E} z$

$x \leq y \leq z$

E is Reflexive, antisymmetric and Transitive.

(d) the domain of Relation D is $\mathbb{Z}$ and $x \, b \, y \Leftrightarrow$ x every divides y

① D is Reflexive

As $x = x$

$\rightarrow x = 1.x$

$\rightarrow x \, D \, x$

② D is not Symmetric and D is not antisymmetric

Suppose $x D y \rightarrow y = x^n$, $n \in \mathbb{Z}$

$\rightarrow x = \frac{1}{n} y$ but $\frac{1}{n} \notin \mathbb{Z}$

$\therefore x D y \rightarrow y D x$

$\therefore$ D is not Symmetric

Also D is not antisymmetric

$\therefore$ let $b D y \rightarrow y D x$

$\rightarrow y = x^{n_1}$ $x = y^{n_2}$ $n_1 n_2 \in \mathbb{Z}$

$\rightarrow x = x^{n_1 n_2}$

$\rightarrow n_1 n_2 = 1$

$\rightarrow n_1 = n_2 = -1$

if $n_1 = n_2 = -1$

$\rightarrow x = -y$

∴ b is not antisymmetric.

6.2.2 b

(b) This relation in also reflexive and transitive as above but for symmetric, since there is one pair who have different weight the relation is not symmetric due to above mentioned reason.

6.2.4. bd

(b) Domain of Relation is $\langle a,b,c \rangle$

$\therefore R = \{(a,b), (b,c), (a,c)\}$

Here $(a,a) \in R$

$\therefore a \not R a \Rightarrow R$ is not reflexive.

now, $a R b$ but $b \not R a$

$\Rightarrow R$ is not Symmetric

Here $a R b, b R c \quad a R c$

$\Rightarrow R$ is Transitive.

(d) Domain is $\langle a,b \rangle$

$R = \{(a,b), (b,a), (a,a), (b,b)\}$

Here $aRa + bRb$

$\Rightarrow$ R is Reflexive

Here $aRb + bRa \Rightarrow$ R is Symmetric

$\Rightarrow$ aRb, bRa then aRa

$\Rightarrow$ R is Transitive.

6.3.1 bd

(b) Domain of R is $\{a,b,c,d\}$

Range of S is $\{a,b,c,d\}$

$RoS: \{a,b,c,d\} \Rightarrow \{a,b,c,d\}$

$(b,c) \in R, (c,d) \in S \Rightarrow (b,d) \in RoS$

$RoS = \{(b,d)\}$

(d) Domain of R is $\{a,b,c,d\}$

Range of R is $\{b,c,d\}$

$RoR: \{a,b,c,d\} \Rightarrow \{b,c,d\}$

$(b,c) \in R, (c,b) \in R \Rightarrow (b,b) \in R \circ R$

$(c,b) \in R, (b,c) \in R \Rightarrow (c,c) \in R \circ R$

$(a,d) \in R, (d,b) \in R \Rightarrow (a,b) \in R \circ R$

$(d,b) \in R, (b,c) \in R \Rightarrow (d,c) \in R \circ R$

$RoR = \{(b,b), (c,c), (a,b), (d,c)\}$

6.3.2 bd

(b) $R_4 \circ R_1 = R_2$

$x \leq y$ and $x = y$

$\Rightarrow x \leq y$

(d) $R_3 \circ R_1 = R_3$

$x \leq y$ and $x < y$

$\Rightarrow x < y$

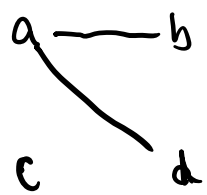
6.3.3 b

(b) $A \circ B = \{(x,y) \mid \exists z$ such that $(x,z) \in B$ and $(z,y) \in A\}$

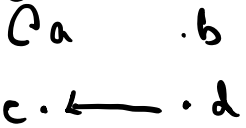
$= \{(a,a), (d,c)\}$

thus

(a) Arrow diagram at $B \circ A$ become



(b) Arrow diagram of $A \circ B$ become



6.3.5 bd

(b) True, Since for all x in D, (x,x) will be in S as

per Reflexive property.

(d) False, suppose S contains (x,y) and (y,x) where $x \neq y$ then S∘S will contains both (x,x) and (y,y) and hence not anti-Reflexive.

6.4.1 bd F H

(b) Out degree of Vertex C is equals to 3

(d) Tail of edge (g,T) is 9

(F) $\langle a,g,f,c,d \rangle$ is not a walk not a Trail as well as not a path.

(H) $\langle c,g,f,e \rangle$ is a not a Circuit because it is not Close and not a Cycle also.

6.4.2 bF

(b) the longest Cycle of the graph is (d,b,c,g,f,d)

(F) No, there is no path of lenght 3 From Vertex a to c.

6.4.3 bdf

(b) 'a' has largest out degree that is 4

(d) Yes, it is a walk and it is open

(f) yes, it is a walk which is Trail but not a path

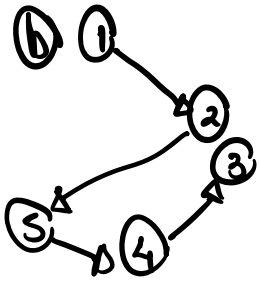
6.5.1 bd

(b) Yes, (b,e) is in G^3; because there is a path of length 3 $(b \to c \to f \to e)$ to reach e from b.

(d) Yes, (g,g) is in G^4; because there is a path of length 4 $(g \to f \to e \to c \to g)$ to reach g from g.

6.5.3 b

(b)



Transitive Form =

	1	2	3	4	5
1	1	1	0	0	0
2	0	1	0	0	1
3	0	0	1	0	0
4	0	0	1	1	0
5	0	0	0	1	1

6.5.4 bh

(b) yes

① → ② → ③ length 2

① → ④ → ③ length 2

④ → ③ → ⑤ length 2

⑤ → ② → ③ length 2

(b) ① → ② → ③ → ①

① → ④ → ③ → ①

yes

there is a walk from vertex 4 to every other vertex in graph.

6.6.2 bd

(b) $A^2 = A.A$

$$A^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \end{bmatrix} = B = [b_{ij}] \ (5.4)$$

$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

(d) From 2nd Column of $A^2 = B$

b_{12}, b_{42}, b_{52} are non zero entries

From v_1 to v_2, v_2 to v_2 or v_2 to v_5 we have walk of length 2

$\therefore v_1, v_4$ and v_5 Can reach Vertex v_2 by walk of length 2

6.6.3 b dF

(b)